

GEORGE ISLAND, MB, Canada

WMO: 714450

Lat: 52.819N

Lon: 97.616W

Elev: 223

StdP: 98.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.7	-29.9	-35.3	0.1	-31.6	-33.2	0.2	-29.7	15.8	-19.0	14.3	-18.9	4.8	320	0.822

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	24.1	19.7	23.0	19.1	21.9	18.6	20.8	23.0	19.9	22.1	19.0	21.4	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.1	22.3	19.0	14.1	21.5	18.0	13.3	20.7	61.5	23.0	58.2	22.2	55.2	21.4	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.6	12.2	-34.3	26.7	3.1	1.1	-36.5	27.5	-38.4	28.1	-40.1	28.8	-42.4	29.5		
(5)			-34.4	21.8	3.0	0.9	-36.5	22.4	-38.3	23.0	-39.9	23.5	-42.1	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.3	-16.9	-9.9	-2.4	4.7	13.1	18.5	18.3	13.5	5.5	-3.7	-13.7	(6)
(7)		DBStd	13.98	7.50	6.95	7.45	5.48	3.89	3.83	2.63	2.66	3.74	3.89	6.07	7.15	(7)
(8)		HDD10.0	4102	877	752	616	371	169	14	0	0	12	147	410	735	(8)
(9)		HDD18.3	6461	1135	986	874	621	423	159	31	34	148	398	660	993	(9)
(10)		CDD10.0	757	0	0	0	0	4	108	263	257	118	7	0	0	(10)
(11)		CDD18.3	75	0	0	0	0	0	4	35	32	4	0	0	0	(11)
(12)		CDH23.3	72	0	0	0	0	0	4	39	27	1	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.0	5.9	5.6	5.6	5.5	5.7	5.1	5.2	6.0	6.8	7.3	7.0	6.1	(14)
(15)	Precipitation	PrecAvg	464	17	11	20	18	44	67	57	62	63	39	28	17	(15)
(16)		PrecMax	612	60	21	102	43	86	149	137	156	134	133	92	46	(16)
(17)		PrecMin	183	1	3	1	0	0	18	3	0	9	2	1	1	(17)
(18)		PrecStd	95	15	6	22	13	23	33	35	40	31	31	20	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.5	5.4	9.4	16.6	23.7	26.0	25.5	22.1	15.1	8.2	2.0	(19)
(20)			MCWB	0.4	0.2	2.7	5.2	10.9	18.1	20.3	20.3	17.9	12.1	6.4	0.9	(20)
(21)		2%	DB	-2.0	-0.6	3.3	6.9	13.2	21.5	24.5	24.0	20.6	13.6	6.4	-0.3	(21)
(22)			MCWB	-2.7	-1.4	1.3	3.6	8.9	17.0	19.8	19.7	17.3	11.2	4.6	-1.0	(22)
(23)		5%	DB	-4.5	-3.2	2.0	5.2	11.3	19.8	23.2	22.9	19.6	12.3	5.2	-1.9	(23)
(24)			MCWB	-5.0	-4.0	0.6	2.6	8.0	15.9	19.3	19.3	16.5	10.0	3.6	-2.6	(24)
(25)		10%	DB	-7.0	-6.3	0.8	3.9	9.8	18.4	22.2	21.9	18.5	11.1	4.0	-3.8	(25)
(26)			MCWB	-7.5	-6.9	-0.2	1.9	7.2	15.3	18.8	18.7	15.6	9.1	2.5	-4.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.2	3.0	5.7	11.7	19.1	22.4	22.0	19.1	13.3	6.7	0.8	(27)
(28)			MCDB	1.2	1.0	4.8	8.8	15.0	22.2	24.4	23.7	21.3	14.7	8.1	1.7	(28)
(29)		2%	WB	-2.6	-1.9	1.6	4.0	9.7	17.9	21.0	20.9	17.9	11.6	5.0	-1.0	(29)
(30)			MCDB	-2.1	-1.1	3.0	6.3	12.4	20.6	23.3	22.9	20.0	13.2	6.1	-0.5	(30)
(31)		5%	WB	-5.0	-4.4	0.7	2.9	8.5	16.8	20.1	20.0	16.9	10.3	3.7	-2.6	(31)
(32)			MCDB	-4.5	-3.6	1.8	4.8	10.9	19.1	22.4	22.2	19.2	12.0	5.0	-2.0	(32)
(33)		10%	WB	-7.5	-7.3	-0.1	2.0	7.4	15.7	19.2	19.2	15.9	9.1	2.7	-4.6	(33)
(34)			MCDB	-7.1	-6.7	1.0	3.8	9.6	18.0	21.6	21.4	18.3	10.9	4.0	-4.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	8.6	9.3	8.0	6.3	5.8	4.9	4.4	3.8	4.3	6.0	(35)	
(36)			MCDBR	8.5	9.7	8.3	8.0	8.7	7.4	5.9	5.3	5.0	4.7	3.6	6.1	(36)
(37)			MCWBR	8.4	9.7	6.9	5.5	5.1	4.0	3.4	3.3	3.9	3.6	3.1	5.7	(37)
(38)		5% WB	MCDBR	8.7	10.2	7.7	7.4	7.8	6.3	5.4	4.9	4.7	4.2	3.6	6.1	(38)
(39)			MCWBR	8.4	9.7	6.4	5.2	5.0	3.9	3.6	3.4	3.9	3.6	3.2	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.255	0.254	0.308	0.346	0.354	0.371	0.387	0.363	0.329	0.295	0.252	0.232	(40)
(41)		taud		2.211	2.280	2.253	2.358	2.429	2.394	2.375	2.472	2.526	2.567	2.363	2.255	(41)
(42)		Ebn at Noon		735	872	879	888	897	880	858	862	857	825	753	708	(42)
(43)		Edh at Noon		64	82	105	109	107	112	112	97	81	63	55	50	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.87	1.86	3.16	4.70	5.49	5.74	5.67	4.68	3.25	1.76	0.89	0.64	(44)
(45)		RadStd		0.09	0.14	0.30	0.40	0.42	0.41	0.29	0.34	0.35	0.20	0.09	0.09	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page